Homework 3: Migration Trends and Selection

Jacob Herbstman

April 23, 2026

1. How has cross-state annual U.S. migration evolved between 2000 and 2024 for individuals aged 25-54 born in the United States, separately by education group?

   (i)

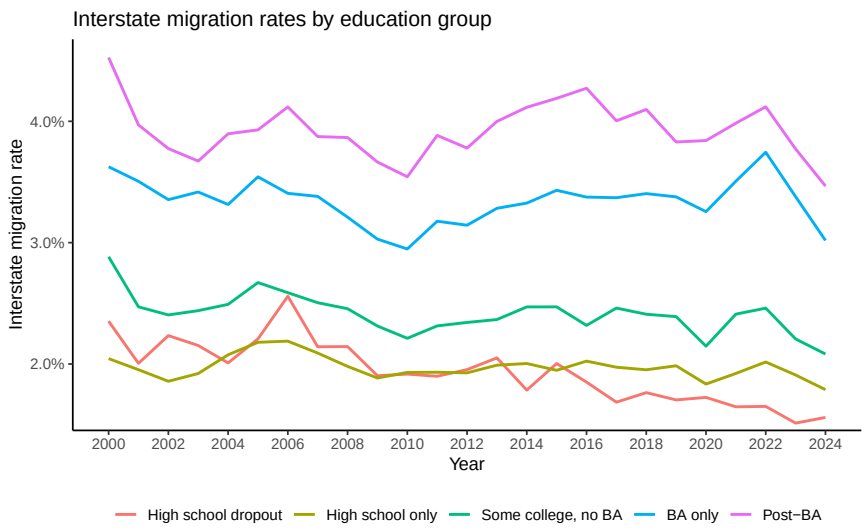

Figure 1: Interstate migration rates by education group, 2000 to 2024. Rates are weighted shares of U.S.-born household residents ages 25 to 54.

   (ii) Every education group migrates at lower rates in 2024 than in 2000. The high-school-only rate falls from 2.04% to 1.79%, and the post-BA rate falls from 4.53% to 3.47%. More educated workers move more in every year, but all five series drift down together.

   (iii) The pattern points to a broad mobility decline, not a change confined to one part of the education distribution. That pushes me toward explanations that affect moving costs or moving returns for many workers at once, rather than a story specific to one skill group.

2. What are some possible reasons why inter-state mobility has declined, and how would I test those theories?

   (i) Housing costs and supply restrictions make it harder to move into high-productivity places (Ganong and Shoag 2017). One test is to interact migration rates with local house-price growth, rent growth, and Saiz-style supply elasticity, then check whether migration has become more sensitive to housing costs over time.

   (ii) The return to moving for work may have fallen for two related reasons. The income gains from moving may have fallen because locations have become more similar on observable and unobservable dimensions (Kaplan and Schulhofer-Wohl 2017), and job-to-job transitions may have slowed (Molloy, Smith and Wozniak 2017). To test this mechanism, I would compare the migration response to local employment shocks across decades and check whether migration rates move with job-changing rates.

   (iii) Household ties and place-specific amenities may be a third factor. There is substantial evidence that people do not leave depressed places at the rate they "should", suggesting local ties or place-specific amenities keeping them there (Zabek 2024). I would use state of birth, the closest proxy available in ACS, and check whether the share of people still living in their birth state has risen over time. A cleaner alternative within these data is to compare migration trends for homeowners versus renters, since homeownership is a standard proxy for rootedness.

   (iv) For a full test, I would merge the ACS repeated cross-sections with local labor-market and housing data, so these mechanisms can be tested in the same regression framework. The decline here is consistent with the earlier evidence in (Molloy, Smith and Wozniak 2011).

3. Define the top 75 largest MSAs in the 2006 sample for the eligible population.

   (i) The sample is individuals ages 25 to 54, born in the United States, living in households. I exclude group quarters.

   (ii) I rank MSAs by weighted 2006 population using current MSA of residence and keep the top 75. The five largest are New York, Chicago, Los Angeles, Philadelphia, and Dallas-Fort Worth.

   (iii) The top-75 list is frozen at 2006 and used as the focal set for all later MSA comparisons.

4. For each of the 75 large MSAs, define total population, inflows, and outflows for 2006, 2010, and 2019.

   (i) Population is the weighted count of the focal sample in the MSA. Inflows are movers currently in the focal MSA whose prior MSA is identifiable and different. Outflows are movers whose prior MSA is the focal MSA

and whose current MSA is identifiable and different. The focal set is the 2006 top 75 MSAs, but the non-focal side of the move can be any identifiable MSA in the ACS, not only another top-75 MSA. Moves involving abroad, non-MSA, or unidentified origins or destinations are excluded from these strict MSA-to-MSA flow counts. If a focal MSA has no positive prior-MSA support in a year, I leave outflows and both flow rates missing rather than setting them to zero.

(ii) The 2010 and 2019 versions use the same 2006 top-75 list and the same definitions.

5. Which MSAs had the highest gross flow rates and highest net flow rates in 2006?

(i)

MSA	Gross flow rate
New Orleans-Metairie, LA	0.236
Orlando-Kissimmee-Sanford, FL	0.118
Las Vegas-Henderson-Paradise, NV	0.113
San Diego-Carlsbad, CA	0.109
Urban Honolulu, HI	0.107
Tucson, AZ	0.099
Austin-Round Rock, TX	0.098
Miami-Fort Lauderdale-West Palm Beach, FL	0.094
San Jose-Sunnyvale-Santa Clara, CA	0.091
San Francisco-Oakland-Hayward, CA	0.088

Table 1: Top 10 gross flow rates in 2006. New Orleans is a Katrina-displacement outlier and should not be read as a normal high-churn case.

Notes: Sample is the top 75 MSAs by weighted 2006 population among U.S.-born household residents ages 25 to 54. Gross flow rate is (inflows + outflows)/population. The non-focal side of a move may be any identifiable MSA. Flows exclude abroad, non-MSA, and unidentified origins or destinations. Rankings require measurable prior-MSA origin support for the focal MSA-year.

(ii)

MSA	Net flow rate
Las Vegas-Henderson-Paradise, NV	0.028
Baton Rouge, LA	0.028
Charlotte-Concord-Gastonia, NC-SC	0.023
Portland-Vancouver-Hillsboro, OR-WA	0.019
Jacksonville, FL	0.019
Austin-Round Rock, TX	0.018
Houston-The Woodlands-Sugar Land, TX	0.017
Tucson, AZ	0.016
Knoxville, TN	0.015
Atlanta-Sandy Springs-Roswell, GA	0.015

Table 2: Top 10 net flow rates in 2006.

Notes: Net flow rate is (inflows − outflows)/population using ACS person weights. Inflows and outflows use the strict identifiable MSA-to-MSA definition, with the top-75 list used only to choose focal MSAs. Rankings require nonmissing outflows and flow rates.

(iii) New Orleans tops the gross-flow table because Katrina outflows were still enormous in 2006. Baton Rouge is also likely affected by Katrina-related displacement, so I do not interpret its 2006 net inflow as a normal high-growth-metro result. New Orleans' gross flow rate is 0.236, but its net flow rate is −0.193. Outside the Katrina-affected entries, the 2006 lists mostly pick up high-turnover and growing Sun Belt or coastal metros.

6. Which MSAs had the largest increase and largest decrease in net flow rates between 2006 and 2010?

(i)

MSA	Change in net flow rate
New Orleans-Metairie, LA	0.193
Miami-Fort Lauderdale-West Palm Beach, FL	0.019
Omaha-Council Bluffs, NE-IA	0.017
Dayton, OH	0.016
Worcester, MA-CT	0.013
San Jose-Sunnyvale-Santa Clara, CA	0.011
Virginia Beach-Norfolk-Newport News, VA-NC	0.010
Orlando-Kissimmee-Sanford, FL	0.009
Greensboro-High Point, NC	0.009
Los Angeles-Long Beach-Anaheim, CA	0.009

Table 3: Top 10 increases in net flow rates, 2006 to 2010.

Notes: Change is the 2010 net flow rate minus the 2006 net flow rate. The focal set is the 2006 top 75 MSAs. The non-focal side of a move may be any identifiable MSA; abroad, non-MSA, and unidentified origins or destinations are excluded. Rankings require nonmissing flow rates in both years.

(ii)

MSA	Change in net flow rate
Las Vegas-Henderson-Paradise, NV	-0.028
Baton Rouge, LA	-0.025
Winston-Salem, NC	-0.021
Jacksonville, FL	-0.018
Austin-Round Rock, TX	-0.014
Houston-The Woodlands-Sugar Land, TX	-0.014
Knoxville, TN	-0.014
Atlanta-Sandy Springs-Roswell, GA	-0.014
Milwaukee-Waukesha-West Allis, WI	-0.014
Grand Rapids-Wyoming, MI	-0.014

Table 4: Top 10 decreases in net flow rates, 2006 to 2010.

Notes: Change is the 2010 net flow rate minus the 2006 net flow rate. Negative values indicate a decline in net inflows relative to population. The focal set is the 2006 top 75 MSAs; flow counts use only strict identifiable MSA-to-MSA moves. Rankings require nonmissing flow rates in both years.

(iii) New Orleans leads the increase list because 2006 is an abnormal base year. I would not read that entry as ordinary recovery-period growth. After New Orleans, the increases are mixed across Florida, California, and older eastern metros. During the downturn, a higher net-flow rate can reflect weaker outflows rather than stronger inflows or a healthier local labor market. The decreases are concentrated in former growth or high-churn metros, with some older industrial metros also appearing in the rankings.

7. Which MSAs had the largest increase and largest decrease in net flow rates between 2006 and 2019?

(i)

MSA	Change in net flow rate
New Orleans-Metairie, LA	0.198
Dayton, OH	0.027
Allentown-Bethlehem-Easton, PA-NJ	0.024
Fresno, CA	0.016
Miami-Fort Lauderdale-West Palm Beach, FL	0.015
Birmingham-Hoover, AL	0.014
Omaha-Council Bluffs, NE-IA	0.014
Toledo, OH	0.014
Albany-Schenectady-Troy, NY	0.013
Sacramento–Roseville–Arden-Arcade, CA	0.009

Table 5: Top 10 increases in net flow rates, 2006 to 2019.

Notes: Change is the 2019 net flow rate minus the 2006 net flow rate. The focal set is the 2006 top 75 MSAs. The non-focal side of a move may be any identifiable MSA; abroad, non-MSA, and unidentified origins or destinations are excluded. Rankings require nonmissing flow rates in both years.

(ii)

MSA	Change in net flow rate
Baton Rouge, LA	-0.028
Albuquerque, NM	-0.026
Bridgeport-Stamford-Norwalk, CT	-0.019
Little Rock-North Little Rock-Conway, AR	-0.018
Milwaukee-Waukesha-West Allis, WI	-0.017
Las Vegas-Henderson-Paradise, NV	-0.017
Portland-Vancouver-Hillsboro, OR-WA	-0.016
Scranton–Wilkes-Barre–Hazleton, PA	-0.016
San Antonio-New Braunfels, TX	-0.015
Grand Rapids-Wyoming, MI	-0.014

Table 6: Top 10 decreases in net flow rates, 2006 to 2019.

Notes: Change is the 2019 net flow rate minus the 2006 net flow rate. Negative values mean the MSA's net flow rate fell over the period. The ranking uses strict identifiable MSA-to-MSA flow counts and requires nonmissing flow rates in both years.

(iii) By 2019, New Orleans dominates because 2006 is an abnormal base year. The largest increases include Dayton, Allentown, and Fresno, where net flows improved relative to the 2006 baseline. On the negative side, the list includes former growth or high-churn metros as well as older industrial metros.

8. For each of the 75 large MSAs, define employment rates for residents, in-migrants, and out-migrants for 2005, 2010, and 2019.

   (i) In 2005, the pooled weighted resident employment rate across the 75

MSAs is 0.786. The pooled weighted in-migrant employment rate is 0.713 and the pooled weighted out-migrant rate is 0.695, so movers start the period below residents.

(ii) In 2010, resident employment falls to 0.770, a 1.6 percentage point decline. In-migrant employment falls 1.7 percentage points to 0.696, and out-migrant employment falls 1.4 percentage points to 0.681. Mover and resident employment rates fall by similar amounts during the recession.

(iii) In 2019, the pooled weighted resident employment rate recovers to 0.829. In-migrant employment recovers to 0.813, while out-migrant employment recovers to 0.791. Movers remain less employed than residents, but the gap is smaller than in 2005.

(iv) Employment rates are weighted shares employed among people with valid employment-status responses. For movers, I count anyone tied to the focal MSA who moved in the last year, excluding only people explicitly observed in the same MSA one year earlier. I use the broader mover sample here because employment is observed even when one endpoint of the MSA move is not identifiable. The strict MSA-to-MSA restriction is needed for the flow-rate accounting exercise. The output folder also contains the full 75-MSA-by-year employment-rate table.

9. Create and interpret the two-panel scatter plot relating changes in resident employment rates to changes in mover employment rates between 2005 and 2010.

(i)

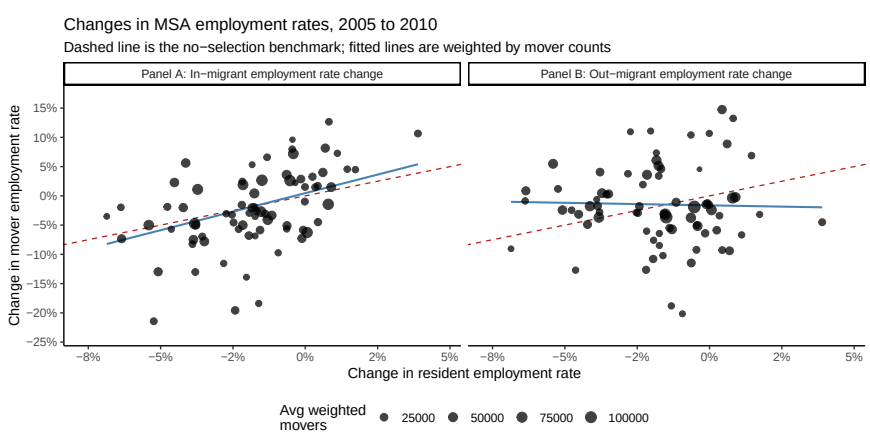

Figure 2: Changes in resident and mover employment rates, 2005 to 2010. Each point is one of the top 75 MSAs. The dashed $y = x$ line is the no-selection benchmark.

Panel	Slope	SE	R^2	$p(\beta = 1)$
A: In-migrants	1.267	0.305	0.191	0.385
B: Out-migrants	-0.087	0.330	0.001	0.002

Table 7: Weighted regressions of mover employment-rate changes on resident employment-rate changes. Weights are the average weighted mover populations in 2005 and 2010.

Notes: Each observation is one top-75 MSA. Changes are 2010 minus 2005 employment rates. The last column reports a two-sided test of the slope against 1.

Under the no-selection benchmark in the prompt, mover employment-rate changes should track resident employment-rate changes for the focal MSA. For in-migrants, the focal MSA is the destination; for out-migrants, it is the origin. A slope of one therefore means that mover employment changes one-for-one with resident employment.

In Panel A the slope is 1.267 (SE 0.305). I cannot reject the no-selection benchmark of one ($p = 0.385$). The estimate is consistent with in-migrant employment rates co-moving roughly one-for-one with resident rates across MSAs, but the standard error leaves room for modest selection in either direction.

(ii) In Panel B the slope is -0.087 (SE 0.330). I reject one ($p = 0.002$). Out-migrant employment rates do not track local conditions. When an MSA's resident employment falls, out-migrants' employment rates stay roughly where they were.

(iii) The selection evidence is mostly on the out-migration side. In-migrants behave close to the no-selection benchmark. Out-migrants don't. Their employment rate is unresponsive to local conditions, meaning the people who leave a downturn MSA look more employed than a random draw of that MSA's population. That is a cross-MSA result, not the pooled average in Q8: average resident and mover employment rates fall by similar amounts, but the out-migrant drop is not larger in MSAs where resident employment fell more. One interpretation is that employed workers are more likely to have outside options and therefore are less negatively selected out of declining MSAs, while lower-employment residents are more likely to remain. That is consistent with evidence that U.S. natives reallocate weakly across MSAs in response to local demand shocks (Cadena and Kovak 2016).

References

Cadena, Brian C. and Brian K. Kovak, "Immigrants Equilibrate Local Labor Markets: Evidence from the Great Recession," *American Economic Journal: Applied Economics*, January 2016, *8* (1), 257–290.

Ganong, Peter and Daniel Shoag, "Why has regional income convergence in the U.S. declined?," *Journal of Urban Economics*, November 2017, *102*, 76–90.

Kaplan, Greg and Sam Schulhofer-Wohl, "Understanding The Long-Run Decline in Interstate Migration," *International Economic Review*, February 2017, *58* (1), 57–94.

Molloy, Raven, Christopher L Smith, and Abigail Wozniak, "Internal Migration in the United States," *Journal of Economic Perspectives*, August 2011, *25* (3), 173–196.

____, Christopher L. Smith, and Abigail Wozniak, "Job Changing and the Decline in Long-Distance Migration in the United States," *Demography*, April 2017, *54* (2), 631–653.

Zabek, Mike, "Local Ties in Spatial Equilibrium," *American Economic Journal: Macroeconomics*, April 2024, *16* (2), 287–317.